

FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10 – SUB-STRAND 2.5: AREA OF PART OF A CIRCLE – FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

Driving question: How can we calculate the exact area of any wedge-shaped or segment-shaped piece of a circle, like a watermelon slice at Wakulima Market, so that we can solve fair-sharing and design problems in real life? Use what you observed, calculated and drew during the whole sequence of lessons to answer each prompt below. Write in full sentences, show every step of your working, and include units (cm^2 or m^2). Where a question gives $r = 7 \text{ cm}$, use $\pi = 22/7$ so that $\pi r^2 = 154 \text{ cm}^2$. For all other questions, use the π button on your calculator and round as the question asks. You may use diagrams and labelled sketches to support your explanations.

Part 1: Why do the pieces look so different?

At Wakulima Market in Nairobi, Mama Wanjiku slices one large, perfectly circular watermelon into wedge-shaped pieces. Customers complain that some pieces look smaller than others even though she says she cut fairly. In class you studied four wedge-shaped sectors, all with radius 7 cm, with central angles of 30° , 90° , 120° and 180° .

(a) Which two measurements decide the size of a watermelon wedge?

(b) Mama Wanjiku says: 'All my pieces have the same radius, so they must be the same size.'

Explain why she is wrong, using the 30° and 180° wedges.

(c) Calculate the area of each of the four wedges ($r = 7 \text{ cm}$). Then state what happens to the area when the angle is doubled from 90° to 180° .

(a) The radius (r) and the central angle (θ), which is the angle between the two straight cuts at the centre of the melon.

(b) Mama Wanjiku is wrong because the radius alone does not fix the area of a wedge. When the radius stays the same, the area changes with the central angle. A 30° wedge is only a thin sliver, while a 180° wedge is half of the whole melon, so the 180° wedge holds 6 times as much watermelon as the 30° wedge ($180 \div 30 = 6$). The customers are right that the pieces are unequal.

(c) Full circle area = $\pi r^2 = 22/7 \times 7 \times 7 = 154 \text{ cm}^2$.

30° : $30/360 \times 154 = 12.83 \text{ cm}^2$ (about 12.8 cm^2).

90° : $90/360 \times 154 = 38.5 \text{ cm}^2$.

120° : $120/360 \times 154 = 51.33 \text{ cm}^2$ (about 51.3 cm^2).

180° : $180/360 \times 154 = 77 \text{ cm}^2$.

When the angle doubles from 90° to 180° , the area doubles from 38.5 cm^2 to 77 cm^2 . This matches what we saw when we counted squares on grid paper. Area is proportional to the central angle.

Part 2: Building the sector area formula

Your class built a formula for the area of a sector by treating a sector as a fraction of a whole circle.

(a) Explain in your own words how you can write the fraction of the circle that a sector with central angle θ takes up. Mention what four 90° sectors do when you put them together.

(b) Use your idea to write a formula for the area of a sector in terms of θ and r .

(c) A wedge is cut from a watermelon with radius 7 cm and has a central angle of 120° . Use your formula to find its area.

(d) Another watermelon is twice as wide in radius as the first one ($r = 14$ cm), and the same 120° wedge is cut. By what factor does the area of the wedge change? Explain why without needing the whole calculation.

(a) A full turn at the centre of a circle is 360° . A sector with central angle θ is therefore $\theta/360$ of the whole circle. For example, four 90° sectors fit together to make the full circle, so each 90° sector is $90/360 = 1/4$ of the circle.

(b) Area of sector = $(\theta/360) \times \pi r^2$, where θ is the central angle in degrees and r is the radius.

(c) $A = (120/360) \times 22/7 \times 7 \times 7 = 1/3 \times 154 = 51.33 \text{ cm}^2$ (about 51.3 cm^2).

(d) The area becomes 4 times as large. Since area depends on r^2 , doubling the radius multiplies πr^2 by $2 \times 2 = 4$, while the fraction $\theta/360$ stays the same. Check: for $r = 14$ cm, $1/3 \times 22/7 \times 14 \times 14 = 205.33 \text{ cm}^2$, which is $4 \times 51.33 \text{ cm}^2$.

Part 3: From a wedge to a curved slice (segment)

Mama Wanjiku now cuts straight across a wedge along a chord, so the piece with the curved rind is a segment, bounded by one chord and one arc.

(a) Draw and label a sector with centre O, radius r and central angle θ . Draw the chord AB and shade the segment. Name the other shape that is inside the sector.

(a) Sketch: a circle centre O, radii OA and OB with angle θ between them, and a straight chord AB. The sector is the whole region OAB bounded by the two radii and the arc. The isosceles triangle OAB (two radii and the chord) sits inside the sector. The segment is the curved region between the chord AB and the arc, and is the part shaded outside the triangle.

(b) The sector is made of the triangle plus the segment, so Segment = Sector – Triangle.

(c) Area of sector = $(\theta/360) \times \pi r^2$. Area of triangle = $\frac{1}{2}r^2 \sin \theta$ (two sides r and r with the included angle θ). So Area of segment = $(\theta/360) \times \pi r^2 - \frac{1}{2}r^2 \sin \theta$.

(d) Sector: $90/360 \times \pi \times 18^2 = 1/4 \times 1017.88 = 254.5 \text{ cm}^2$. Triangle: $\frac{1}{2} \times 18^2 \times \sin 90^\circ = \frac{1}{2} \times 324 \times 1 = 162.0 \text{ cm}^2$. Segment = $254.5 - 162.0 = 92.5 \text{ cm}^2$.

(b) Explain how you can find the area of the segment from the area of the sector.

(c) Write the formula for the area of the segment. Show where each part of the formula comes from.

(d) A watermelon has radius 18 cm. Find the area of the segment cut off by a chord that makes a central angle of 90° . Give your answer to 1 decimal place.

Part 4: The fair-share verdict at Wakulima Market

Mama Wanjiku cut four curved segments from a new watermelon with radius 18 cm. Each piece was cut by a chord that makes the central angle shown below. The customers are Customer A, B, C and D.

Customer	Central angle θ
A	40°
B	65°
C	90°
D	120°

(a) Calculate the sector area, the triangle area and the segment area for each customer. Present your results in a table (1 decimal place).

(b) Which customer received the largest segment, and which received the smallest?

(a) Full circle area = $\pi \times 18^2 = 1017.9 \text{ cm}^2$. Triangle area = $\frac{1}{2} \times 18^2 \times \sin \theta = 162 \sin \theta$.

Customer	θ	Sector area (cm^2)	Triangle area (cm^2)	Segment area (cm^2)
A	40°	113.1	104.1	9.0
B	65°	183.8	146.8	37.0
C	90°	254.5	162.0	92.5
D	120°	339.3	140.3	199.0

Example working for Customer B: sector = $65/360 \times 1017.9 = 183.8 \text{ cm}^2$; triangle = $162 \times \sin 65^\circ = 146.8 \text{ cm}^2$; segment = $183.8 - 146.8 = 37.0 \text{ cm}^2$.

(b) Customer D (120°) received the largest segment, 199.0 cm^2 . Customer A (40°) received the smallest, 9.0 cm^2 .

(c) Recommendation: The pieces are not equal, so charging one price is unfair. Customer D gets about 22 times as much watermelon as Customer A ($199.0 \div 9.0 \approx 22$). The appearance of a piece cannot be trusted, so Mama Wanjiku should price each piece by its area in cm^2 , found from the central angle θ and radius r using the sector formula (for wedges) or the segment formula (for curved slices). She could also choose the angles beforehand so that every customer's piece has the same calculated area.

(c) Mama Wanjiku charges the same price for every piece. Write a recommendation to her, based on your calculations, explaining how she could price the pieces fairly.	
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Part 5: Using area of part of a circle in real life	
The KICD key inquiry question asks: Where do we use area of part of a circle in real life? (a) A centre-pivot irrigation arm in Naivasha has radius 80 m and sweeps through an angle of 270°. Calculate the area watered, in m² to the nearest whole number, and then in hectares (1 hectare = 10 000 m²) to 2 decimal places. (b) Give one other real-life situation in Kenya where you would calculate the area of a sector or a segment, and say which formula you would use. (c) Describe how your thinking about the driving question has changed since the first lesson, when you drew your first model of the watermelon wedge. Name at least two ideas you added to your model.	(a) This is a sector with $\theta = 270^\circ$ and $r = 80$ m. $A = 270/360 \times \pi \times 80^2 = 0.75 \times 20\,106.2 = 15\,080$ m² (to the nearest whole number). In hectares: $15\,080 \div 10\,000 = 1.51$ hectares. (b) Example: a school gate in Kisumu of radius 1.2 m that swings through 75° sweeps out a sector, so the area it clears is found with $A = (\theta/360) \times \pi r^2$. Other acceptable answers: a door sweep, a slice of ugali cake, or a curved flower bed with a straight edge, which is a segment found using $(\theta/360)\pi r^2 - \frac{1}{2}r^2 \sin \theta$. (c) In Lesson 1 I thought only the radius or the way the piece looked decided its size. Now I know that area depends on both the central angle θ and the radius r. My model now has: the sector formula $A = (\theta/360) \times \pi r^2$, showing that a sector is the fraction $\theta/360$ of the whole circle; the segment formula, with the triangle $\frac{1}{2}r^2 \sin \theta$ taken away from the sector; and the ideas that doubling the angle doubles the area while doubling the radius makes it 4 times as large. Using these formulas, we can measure fair shares exactly instead of guessing.

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Understanding of how angle and radius control area	Clearly explains that sector area is proportional to the central angle and to r^2 , using correct comparisons from the data (such as 30° versus 180°) and the anchoring watermelon context.	States that angle and radius both affect area and gives a mostly correct comparison, but explanation of proportionality or the r^2 effect is incomplete.	Says only that bigger pieces have bigger area, or relies on the look of the pieces without mentioning angle and radius.

Correct use of the sector and segment formulas	Writes both formulas correctly, explains where each part comes from (fraction $\theta/360$ of the circle; sector minus triangle), and substitutes accurately with units.	Uses both formulas with only minor slips in substitution, rounding or units, and gives partial reasoning for where they come from.	Confuses sector and segment, omits the triangle subtraction, or makes major errors in substitution.
Accuracy of calculations and presentation of data	All sector, triangle and segment areas are correct, set out in a clear table with working and units, and the largest and smallest pieces are identified correctly.	Most calculations are correct and clearly set out, with one or two small errors that do not change the conclusions.	Several calculation errors, missing working or units, or a table that is incomplete or inconsistent.
Fair-share verdict and real-life application	Gives a clear, evidence-based recommendation to Mama Wanjiku using the calculated areas, and applies the formulas correctly to another Kenyan context such as irrigation.	Gives a reasonable recommendation and a correct real-life example, though links to the calculated evidence are limited.	Recommendation is vague or not supported by calculations, and the real-life example is missing or incorrect.
Reflection on model revision	Describes specific changes from the Lesson 1 model to the final model, naming at least two new ideas and linking them to the driving question.	Mentions at least one idea added to the model with some link to the driving question.	Gives a general statement of learning without naming specific changes to the model.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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